

Space Time Algebra Framework

Discovering STA

Symmetry: is a transformation that leaves something unchanged

Lorentz transform

$$ct' = \gamma \left(ct - \frac{v}{c} x \right)$$

$$x' = \gamma \left(x - \frac{v}{c} ct \right)$$

$$y' = y$$

$$z' = z$$

$$\tau = ct$$

$$\beta = \frac{v}{c}$$

$\Downarrow$

$$t' = \gamma (t - \beta x)$$

$$x' = \gamma (x - \beta t)$$

$$y' = y$$

$$z' = z$$

Mixing Space and Time

$$\{ \hat{t}, \hat{x}, \hat{y}, \hat{z} \}$$

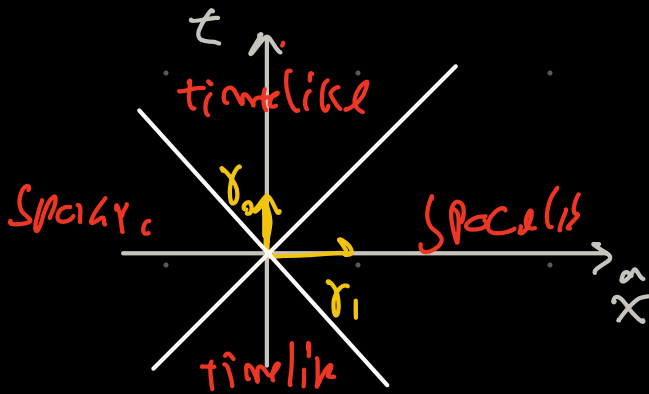
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Conventions Greek letters & dots.

$$\{\gamma_0, \gamma_1, \gamma_2, \gamma_3\}$$

Spacetime vectors a, b, c

Spac vectors $\vec{a}, \vec{b}, \vec{c}$



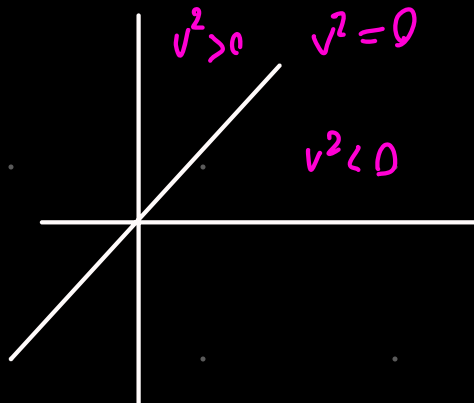
$$|\vec{v}| = \sqrt{\vec{v}^2}$$

Vectors in ST represent events and reference frames

If we want that $V \cdot V = c^2$ in all reference frames

we need that Minkowski space

$$u^2 = a^2 - \beta^2$$



$$t' = \gamma(\alpha - v\beta)$$

$$x' = \gamma(\beta - v\alpha)$$

$$u = \alpha \gamma_0 + \beta \gamma_1$$

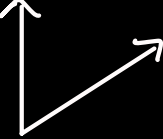
$$u^2 = t'^2 + \cancel{x'^2} = t^2 + x^2 \rightarrow \text{condition}$$

$$\alpha^2 - \beta^2$$

quora @ u r u

transform

$$0 = \gamma(\beta - v\alpha)$$



for a vector
vertical

$$x^1 = 0$$

$$\beta = v\alpha \rightarrow v = \frac{\beta}{\alpha}$$

$$t' = \gamma(\alpha - v^2\alpha)$$

$$u^2 = \gamma^2(\alpha - v\beta)^2$$

$$\frac{1}{1-v^2} (\alpha - v\beta)^2 -$$

$$\frac{1}{1-\frac{\beta^2}{\alpha^2}} \left(\alpha - \frac{\beta^2}{\alpha} \right)^2$$

$$u^2 = \frac{\alpha^2}{\alpha^2 - \beta^2} \left(\frac{\alpha^2 - \beta^2}{\alpha^2} \right)$$

$$u^2 = \alpha^2 - \beta^2$$

Minkowski metric

$$(+, -, -, -) \rightarrow \eta$$

this is the main difference between V&A and STA

$$\gamma_0^2 = 1$$

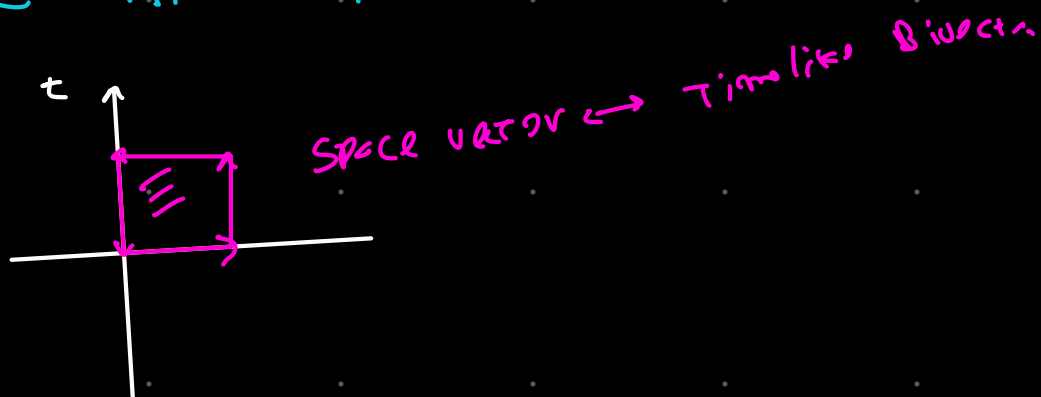
$$\gamma_1^2 = -1$$

$$\gamma_2^2 = -1$$

$$(+, -, -, -)$$

$$\gamma_3^2 = -1$$

SPACE TIME SPLITS



$$\rightarrow V \cdot \gamma_0 \rightarrow \text{Bivector}$$

$$\uparrow V \cdot \gamma_0 \rightarrow \text{Scalar}$$

ST SPLIT

$$\nearrow V \cdot \gamma_0 \rightarrow \text{Bivector} + \text{Scalar}$$

↓ In the γ_0 reference frame -

but we can do it with any reference

ST

γ_0'

SPACE

and the same will just work

$$(\gamma_0 \gamma_0) \longleftrightarrow \text{Scalar}$$

$$\gamma_1 \gamma_0 \longleftrightarrow \hat{x}$$

$$\gamma_2 \gamma_0 \longrightarrow \hat{y}$$

$$\gamma_3 \gamma_0 \longleftrightarrow \hat{z}$$

space and time bivectors

SPACED VECTORS ON STRUCTURES

$$(\gamma_1 \gamma_0)^2 = \gamma_1 \gamma_0 \gamma_1 \gamma_0 = - \gamma_1 \gamma_1 \gamma_0 \gamma_0$$

$$= - \gamma_0 \gamma_0$$

$$= -1$$

www

$$|\hat{x}| = 1$$

testing anti commutativity
 $\rightarrow \hat{x} \hat{y}$

$$\gamma_1 \gamma_0 \gamma_2 \gamma_0 = - \gamma_2 \gamma_0 \gamma_1 \gamma_0$$

$$\hat{y} \hat{x}$$

and what about bivectors in SPACE

$$\begin{matrix} \text{ST} & \text{SPACE} \\ \gamma_1 \gamma_0 \gamma_2 \gamma_0 \rightarrow \gamma_2 \gamma_1 \rightarrow \hat{x} \hat{y} \end{matrix}$$

$$\gamma_1 \gamma_0 \gamma_3 \gamma_0 \rightarrow \gamma_3 \gamma_1 \rightarrow \hat{x} \hat{z}$$

$$\gamma_2 \gamma_0 \gamma_3 \gamma_0 \rightarrow \gamma_3 \gamma_2 \rightarrow \hat{y} \hat{z}$$

$$\gamma_0 \gamma_1 \gamma_2 \gamma_2 \rightarrow \hat{x} \hat{y} z$$

Lorentz Transform

spacetime $\xrightarrow{\text{ST split}}$ space

spacetime $\xrightarrow{\text{Lorentz}}$ spacetime

$$t' = \gamma(t - vx)$$

$$x' = \gamma(x - vt)$$

$$L_v(\alpha \gamma_0 + \beta \gamma_1) = \gamma(\alpha - v\beta) \gamma_0 + (\beta - v\alpha) \gamma_1$$

where starting IRT is γ_0

$$L(\gamma_0') = \gamma_0$$

$$\gamma_0' = \gamma(\gamma_0 + v\gamma_1)$$

$$v \gamma_0' \gamma_0 ?$$

$$(\alpha \gamma_0 + \beta \gamma_1) \gamma_0' \gamma_0 = (\alpha \gamma_0 + \beta \gamma_1) \overbrace{\gamma(\gamma_0 + v\gamma_1)}^{\gamma_0'} \gamma_0$$

↓

is valid

So observe that

this linear
transform

$$L_v(v) = v \gamma_0' \gamma_0$$

maps from
reference frame

where $\gamma_0' = \gamma(\gamma_0 + v \gamma_1)$ $\gamma_0' \rightarrow \gamma_0$

$L_v(v) = v \gamma_0' \gamma_0$ look like a rotation

but a rotation is a linear transformation
that preserves lengths

but in Minkowski space they are

Rotations Lorentz Transform are ST
rotations

n -dimensional Lorentz Boosts

$$\sqrt{\gamma_0 \gamma_0'} \vee \sqrt{\gamma_0' \gamma_0} = v'$$

But look in QED dynamics

$$\nabla F = J \rightarrow \text{vector}$$

↓
Scalar + Vector

↘
Bivector + Vector

but in STA

$$\nabla F = J$$

$$f = q F \cdot v$$

↑
Vector

↘
Bivector

↑
Vector

And the most shocking part

The Dirac Matrices are a matrix

representation of the basis vectors

Dirac equation

$$\nabla \psi \gamma_2 \gamma_1 \hbar - q A \psi = m \psi \gamma_0$$

gradient $\nabla = (\partial_x, \partial_y, \partial_z)$

multivector description

$$\nabla = \partial_x e_x + \partial_y e_y + \partial_z e_z$$

$$\vec{\nabla} (\text{Scalar field function}) \rightarrow \text{gradient}$$

st

$$\vec{\nabla} f = \frac{\partial f}{\partial x} e_x \quad \nabla = \partial_\mu e^\mu?$$

$$\nabla f = \partial_\mu f e^\mu$$

$$\vec{\nabla} \vec{f} =$$

↓

$$\partial_x e_x + \partial_y e_y + \partial_z e_z (E_x \vec{e}_x + E_y \vec{e}_y + E_z \vec{e}_z)$$

$$\partial_x E_x \hat{e}_x \hat{e}_x + \partial_x E_y \hat{e}_x \hat{e}_y + \partial_x E_z \hat{e}_x \hat{e}_z$$

$$\partial_y E_y \hat{e}_y \hat{e}_y + \partial_y E_x \hat{e}_y \hat{e}_x + \partial_y E_z \hat{e}_y \hat{e}_z$$

$$\partial_z E_z \hat{e}_z \hat{e}_z +$$

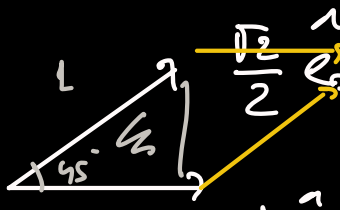
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$$\underbrace{\partial_\mu E^\mu}_{\text{Scalar}} + \underbrace{\nabla \wedge E}_{\text{Vector}} = \vec{\nabla} \cdot \vec{E}$$

$$\nabla \cdot E + \nabla \wedge E = \nabla E$$

$a = (0\hat{e}_0 + 1\hat{e}_1)$ $a \cdot b = 1\hat{e}_1 \hat{e}_0$

 $b = (1\hat{e}_0 + 0\hat{e}_1)$


 $\frac{\sqrt{2}}{2} \hat{e}_0 + \frac{\sqrt{2}}{2} \hat{e}_1$
 $1\hat{e}_0 + 0\hat{e}_1$

$$(1\hat{e}_0 + 0\hat{e}_1) \cdot \left(\frac{\sqrt{2}}{2} \hat{e}_0 + \frac{\sqrt{2}}{2} \hat{e}_1 \right)$$

$$\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \hat{e}_0 \hat{e}_1$$

tem a conservação
de alguma métrica

$$\nabla \cdot E + \underbrace{\nabla \wedge E}_{\text{bivector}} = \nabla E$$

$$\nabla F = \frac{\vec{J}}{c \epsilon_0}$$